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United States Patent Application Publication

20250276523

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

OTSUKA; MANABU

LIQUID EJECTION HEAD SUBSTRATE AND METHOD FOR MANUFACTURING LIQUID EJECTION HEAD SUBSTRATE

Abstract

A technique capable of suppressing the occurrence of damage to a covering layer is provided. A liquid ejection head substrate including an energy generation element for imparting energy to liquid includes a first substrate, a second substrate including a bonded surface bonded to a bonded surface of the first substrate via an adhesive and provided with wiring connected to the energy generation element and a covering layer covering the wiring, and a member provided on the bonded surface of at least one of the first substrate and the second substrate and in a position where the member does not overlap the wiring in a direction in which the first substrate and the second substrate are laminated, the member having a height so that a second thickness of the adhesive on the covering layer provided on the wiring is greater than a first thickness of the adhesive on the member.

Inventors: OTSUKA; MANABU (Kanagawa, JP)

Applicant: CANON KABUSHIKI KAISHA (Tokyo, JP)

Family ID: 96881744

Appl. No.: 19/068611

Filed: March 03, 2025

Foreign Application Priority Data

JP 2024-031791

Mar. 04, 2024

Publication Classification

Int. Cl.: B41J2/14 (20060101); B41J2/16 (20060101)

U.S. Cl.:

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to a liquid ejection head substrate and a method for manufacturing the liquid ejection head substrate.

Description of the Related Art

[0002] As functional devices, such as microfluidic devices, there are known devices formed of a substrate assembly in which substrates are bonded together with an adhesive. Examples of such devices include a liquid ejection head, such as a print head that ejects ink to perform printing.

[0003] Japanese Patent Laid-Open No. 2013-091272 discloses a liquid ejection apparatus including a liquid ejection head with a substrate assembly. In Japanese Patent Laid-Open No. 2013-091272, the substrate assembly includes a pressure generation chamber communicating with a nozzle opening, a piezoelectric layer, and a piezoelectric element having an electrode provided in the piezoelectric layer. In the liquid ejection head, the piezoelectric element is driven to impart pressure to liquid stored in the pressure generation chamber via the piezoelectric layer, thereby ejecting the liquid from the nozzle opening.

[0004] However, in the substrate assembly of Japanese Patent Laid-Open No. 2013-091272, in bonding the bonded surface of one substrate on which wiring connected to the piezoelectric element and a covering layer protecting the wiring are formed to the other substrate with an adhesive, the other substrate may contact the covering layer due to a load during bonding, which may damage the covering layer. Even in a case where the other substrate does not contact the covering layer, stress may be applied to the covering layer due to warpage of the substrates during the bonding step, expansion and contraction of a member due to temperature changes during the step of manufacturing the liquid ejection head, and the like, which may damage the covering layer in a place where the adhesive on the covering layer is thin.

SUMMARY OF THE INVENTION

[0005] The present invention is made in view of the above problems and provides a technique capable of suppressing the occurrence of damage to a covering layer.

[0006] A liquid ejection head substrate includes: an energy generation element for imparting energy to liquid, the liquid ejection head substrate includes a first substrate, a second substrate comprising a bonded surface bonded to a bonded surface of the first substrate via an adhesive and provided with wiring connected to the energy generation element and a covering layer covering the wiring, and a member provided on the bonded surface of at least one of the first substrate and the second substrate and in a position where the member does not overlap the wiring in a direction in which the first substrate and the second substrate are laminated, wherein the member has a height so that a second thickness of the adhesive on the covering layer provided on the wiring is greater than a first thickness of the adhesive on the member.

[0007] The present invention makes it possible to suppress the occurrence of damage to a covering layer.

[0008] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIGS. **1A** and **1B** are cross-sectional configuration diagrams of a liquid ejection head substrate;

[0010] FIGS. **2A** and **2B** are configuration diagrams of the main portion of a liquid ejection head according to the known art;

[0011] FIGS. **3A** and **3B** are diagrams showing the conditions and results of a simulation conducted by the inventor of the present application;

[0012] FIG. **4** is a diagram showing the results of an experiment conducted by the inventor of the present application;

[0013] FIGS. **5A** to **5C** are diagrams for explaining arrangements of spacers on the substrate;

[0014] FIG. **6** is a diagram showing the thickness of an adhesive near a spacer during bonding;

[0015] FIGS. **7A** and **7B** are diagrams showing the positional relationship between a recess and the spacer that may cause a crack in the substrate;

[0016] FIGS. **8A** to **8D** are diagrams showing the positional relationship between the recess and the spacer that suppresses the occurrence of a crack in the substrate;

[0017] FIGS. **9A** to **9C** are diagrams showing the positional relationship between the recess and the spacer that suppresses the occurrence of a crack in the substrate;

[0018] FIGS. **10A** to **10D** are diagrams showing the step of manufacturing the liquid ejection head substrate;

[0019] FIGS. **11A** to **11E** are diagrams showing the step of manufacturing the liquid ejection head substrate;

[0020] FIGS. **12A** to **12C** are diagrams showing the step of manufacturing the liquid ejection head substrate;

[0021] FIGS. **13A** to **13D** are diagrams showing the step of manufacturing the liquid ejection head substrate; and

[0022] FIGS. **14A** and **14B** are perspective configuration diagrams of a liquid ejection apparatus and the liquid ejection head.

DESCRIPTION OF THE EMBODIMENTS

[0023] A description will be given below of an example of embodiments for a liquid ejection head substrate and a method for manufacturing the liquid ejection head substrate with reference to the accompanying drawings. It should be noted that the following embodiments do not limit the present invention, and not all of the combinations of features described in the present embodiment are essential to a solution to the problems to be solved by the present invention. Further, the positions, shapes, and the like of constituent elements described in the embodiments are merely examples and are not intended to limit the present invention only thereto.

Configuration of Liquid Ejection Head Substrate

[0024] First, the configuration of a liquid ejection head substrate will be described. It should be noted that in the present embodiment, a case where three substrates are bonded together as a liquid ejection head substrate will be described as an example, but the present invention is not limited to this and can be applied to a case where two or four or more substrates are bonded together. Further, in the present embodiment, the bonding between wafers is described as an example, but the present invention is not limited to this and can be applied to a case where cut pieces (chips) are bonded to a wafer, or a case where the cut pieces are bonded together.

[0025] FIG. **1A** is a cross-sectional configuration diagram of the liquid ejection head substrate, and FIG. **1B** is an enlarged view of the inside of a frame **1B** in FIG. **1A**. A liquid ejection head substrate **10** has a configuration for ejecting liquid. Specifically, the liquid ejection head substrate **10** includes a pressure chamber (hereinafter also referred to as “liquid chamber”) **20** for storing liquid,

and a flow path **30** for supplying the pressure chamber **20** with liquid. The liquid ejection head substrate **10** also includes a piezoelectric element **26** as an energy generation element for imparting energy to liquid stored in the pressure chamber **20**, and an ejection port **32** through which the liquid stored in the pressure chamber **20** and to which energy is imparted by the piezoelectric element **26** is ejected.

[0026] The liquid ejection head substrate **10** includes a first substrate **12**, a second substrate **14**, and a third substrate **16**, and the above-mentioned configuration for ejecting liquid is formed in a bonded substrate in which the three substrates are laminated and bonded together. One surface **12a** of the first substrate **12** is bonded to one surface **14a** of the second substrate **14** via an adhesive **18**. Further, one surface **16a** of the third substrate **16** is bonded to the other surface **14b** of the second substrate **14** via the adhesive **18**. Hereinafter, the “liquid ejection head substrate” will be referred to as “bonded substrate” as appropriate.

[0027] The first substrate **12** and the second substrate **14** are, for example, silicon substrates. The second substrate **14** is provided with an opening **22** that forms a pressure chamber **20** in a case where the substrates are bonded together, and a vibration film **24** that serves as a ceiling wall of the pressure chamber **20** and divides a plurality of the pressure chambers **20**. The vibration film **24** is provided with a piezoelectric element **26** in a position facing the opening **22** via the vibration film **24**. The liquid ejection head substrate **10** is provided with, for example, the plurality of pressure chambers **20**, and in the present embodiment, the plurality of pressure chambers **20** are provided in a direction orthogonal to a paper surface in FIG. **1A**. Thus, in the second substrate **14**, a plurality of the openings **22** are provided in the direction, and the vibration film **24** is provided with the piezoelectric element **26** in each of positions facing the respective openings **22** via the vibration film **24**.

[0028] The first substrate **12** is provided with a piezoelectric element accommodation portion **28** accommodating the piezoelectric element **26** on the vibration film **24**, and a flow path **30** for supplying the pressure chamber **20** with liquid and collecting liquid from the pressure chamber **20**. The bonded substrate **10** is provided with an accommodation portion (not shown) accommodating liquid on the other surface **12b** side of the first substrate **12**. The liquid accommodated in the accommodation portion is then supplied to the pressure chamber **20** through the flow path **30**.

[0029] The third substrate **16** is, for example, a silicon substrate, and is provided with a plurality of the ejection ports **32**. The ejection ports **32** are provided in positions corresponding to the respective openings **22** (pressure chambers **20**), and in the present embodiment, the plurality of ejection ports **32** are arranged in the direction orthogonal to the paper surface in FIG. **1A**. As a result, in a case where energy is imparted to liquid stored in each pressure chamber **20** by driving the piezoelectric element **26**, the liquid is ejected from the ejection ports **32** to the outside of the bonded substrate **10**.

[0030] In the second substrate **14**, an insulation film **34**, such as a TEOS film, is formed on the one surface **14a**, and wiring **36** connected to the piezoelectric element **26** is formed on the insulation film **34**. A covering layer **38** for protecting the wiring **36** is provided on the insulation film **34** and the wiring **36**. This covering layer **38** is formed in a predetermined region including the wiring **36**, and in the present specification, a portion of the covering layer **38** that covers the wiring **36** is appropriately referred to as a covering portion covering the wiring **36** or the covering layer **38** on the wiring **36**. Further, on the covering layer **38**, there is provided a spacer (hereinafter, also simply referred to as “member”) **40** for ensuring the thickness of the adhesive **18** between the covering layer **38** on the wiring **36** and the one surface **12a** of the first substrate **12**.

[0031] In the present specification, unless otherwise specified, the term “thickness” refers to a length in a direction in which the substrates are laminated in the bonded substrate **10** (the vertical direction in FIGS. **1A** and **1B**). In the following description, “a direction in which the substrates are laminated in the bonded substrate **10**” is simply referred to as “laminated direction.” In the present embodiment, the one surface **12a** of the first substrate **12** is a bonded surface bonded to the second

substrate **14**, and the one surface **14a** side of the second substrate **14** on which the above-mentioned functional film is formed is a bonded surface bonded to the first substrate **12**. That is, the bonded surface of the second substrate **14** bonded to the first substrate **12** includes the above-mentioned functional film with the one surface **14a**. Incidentally, in the second substrate **14**, the other surface **14b** is the other bonded surface that faces one bonded surface bonded to the first substrate **12** and is bonded to the third substrate **16**.

[0032] In the bonded substrate **10**, a piezoelectric actuator portion **42** including the piezoelectric element **26** and the like is formed adjacent to the pressure chamber **20**. A lower electrode (not shown) connected to the piezoelectric element **26** between the piezoelectric element **26** and the vibration film **24** and an upper electrode (not shown) connected to a piezoelectric element on the piezoelectric element **26** are formed on the vibration film **24** serving as the ceiling wall of the pressure chamber **20**. The vibration film **24** is formed by, for example, plasma CVD. A hydrogen barrier film (not shown), the lower electrode, the piezoelectric element **26**, and the upper electrode are formed on the vibration film **24** in this order.

[0033] The lower electrode and the upper electrode are formed by, for example, the sputtering method, and the piezoelectric element **26** is formed by the sol-gel method. The piezoelectric element **26** may be formed by the sputtering method. For example, a PZT (lead zirconate titanate) film formed by the sol-gel method or the sputtering method can be used for the piezoelectric element **26**. The piezoelectric element **26** is made of a sintered body of metal oxide crystal. On the one surface **14a** side of the second substrate **14**, an interlayer film and the wiring **36** are formed so that the piezoelectric element **26** can be driven, and the piezoelectric actuator portion **42** is configured to be drivable.

[0034] The piezoelectric element **26** is provided in a position facing the pressure chamber **20** via the vibration film **24**. The vibration film **24** has deformability. The wiring **36** is connected to a pad **44** for electrical connection which serves as a connection terminal for connection with the outside, and is connected to the piezoelectric element **26** via an upper electrode and a lower electrode. In a case where a driving voltage is applied to the piezoelectric element **26** from an external driving IC (not shown) via the pad **44**, the wiring **36**, the upper electrode, and the lower electrode, the piezoelectric element **26** is deformed by a reverse voltage effect. The deformation of the piezoelectric element **26** deforms the vibration film **24**, which results in a change in the volume of the pressure chamber **20**, and liquid stored in the pressure chamber **20** is pressurized. The liquid is then ejected from each ejection port **32** communicating with the pressure chamber **20** as droplets by this pressurization, to the outside of the bonded substrate **10**.

Concerns That may Arise in a Case Where no Spacer is Provided

[0035] As described above, the bonded substrate **10** is provided with the spacer **40** for ensuring the thickness of the adhesive **18** between the covering layer **38** on the wiring **36** and the one surface **12a** of the first substrate **12**. Here, concerns that may arise in a case where the spacer **40** is not provided will be described with reference to FIGS. 2A and 2B.

[0036] FIGS. 2A and 2B are diagrams for explaining concerns that may arise in a case where the spacer **40** is not provided. FIG. 2A is an enlarged view of the vicinity of the wiring **36** in a portion where the first substrate **12** and the second substrate **14** are adhered to each other. FIG. 2B is an enlarged view of the vicinity of the pad **44** connected to an external terminal **200** and sealed with a sealing agent **202**.

[0037] In a case where the spacer **40** is not provided between the first substrate **12** and the second substrate **14**, a load during bonding may cause contact between the one surface **12a** and the covering layer **38** on the wiring **36** in a case where the first substrate **12** and the second substrate **14** are bonded together via the adhesive **18** (see FIG. 2A). In this case, the load during bonding may be directly applied to the covering layer **38** formed on the wiring **36**, which may damage the covering layer **38**. Even in a place where the covering layer **38** on the wiring **36** and the one surface **12a** are close to each other but not in contact with each other, there is a possibility that the covering layer

38 is damaged near an end of the substrate due to stress caused by warpage of the substrate during the manufacturing step or expansion and contraction of the sealing agent that seals the connection portion connected with the external terminal (see FIG. 2B).

[0038] Such damage to the covering layer **38** is likely to occur, for example, in a case where the wiring **36** is made of a relatively soft metal such as a metal containing aluminum and where the covering layer **38** is formed of a hard film such as a Si compound (silicon compound) film. That is, the wiring **36** made of a soft metal can be deformed by the above-mentioned stress, while the covering layer **38** formed of a hard film is damaged in the process of deformation due to the above-mentioned stress. Further, in a case where the relationship between the thickness **T1** of the wiring **36** made of a soft metal and the thickness **T2** of the covering layer **38** formed of a hard film is $T1 > T2$, the covering layer **38** cannot be deformed in response to the deformation of the wiring **36**, which increases the possibility of the occurrence of damage.

Verification by the Inventor of the Present Application

[0039] Here, a description will be given of the results of a simulation of compressive stress on the covering layer **38** and an experiment using a temperature cycle test conducted by the inventor of the present application. FIGS. 3A and 3B are diagrams for explaining the simulation conducted by the inventor of the present application. FIG. 3A is a diagram showing the configuration of a substrate on which the simulation was conducted. FIG. 3B is a graph showing the results of the simulation. FIG. 4 is a diagram showing the results of the experiment conducted by the inventor of the present application to verify the crack occurrence rate in the covering layer **38** by conducting the temperature cycle test.

[0040] The substrate on which this simulation was conducted imitated the bonded substrate **10** by using Si as the first substrate **12** and the second substrate **14**, benzocyclobutene as the adhesive **18**, AlCu as the wiring **36**, and SiN as the covering layer **38** (see FIG. 3A). Further, a resin was used as the sealing agent **202**. The compressive stress applied to the covering layer **38** in a region A (see FIG. 3A) which is a bonded portion between the first substrate **12** and the second substrate **14** and is adjacent to the sealing agent **202** in a case where the temperature was changed from 130° C. to -40° C. was simulated for the substrate imitating the bonded substrate **10**. In this simulation, the above-mentioned compressive stress was obtained for a plurality of samples in which the thickness of the adhesive **18** bonding the first substrate **12** and the second substrate **14** together differs.

[0041] The results are shown in FIG. 3B. In FIG. 3B, the horizontal axis represents the thickness of the adhesive **18**, and the vertical axis represents the compressive stress. As shown in FIG. 3B, the smaller the thickness of the adhesive **18** that bonds the first substrate **12** and the second substrate **14** together is, the larger the compressive stress becomes, and this tendency is more pronounced in a case where the thickness is less than 0.2 μm .

[0042] Further, the temperature cycle test was actually conducted using the above-mentioned substrate imitating the bonded substrate **10** to check whether there was a crack in the covering layer **38** according to the thickness of the adhesive **18** between the covering layer **38** on the wiring **36** and the one surface **12a** of the first substrate **12**. In this temperature cycle test, the high temperature was 130° C., and the low temperature was -40° C. The results of this temperature cycle test are shown in FIG. 4. In a case where the thickness of the adhesive **18** was less than 100 nm, cracks occurred in the covering layer **38** in all three samples, and the crack occurrence rate was 100%. In a case where the thickness of the adhesive **18** was 100 nm or more and less than 200 nm, cracks occurred in the covering layer **38** in two samples out of four samples, and the crack occurrence rate was 50%. Further, in a case where the thickness of the adhesive **18** was 200 nm or more and less than 300 nm, no cracks occurred in the covering layer **38** in all three samples, and the crack occurrence rate was 0%. Furthermore, in a case where the thickness of the adhesive **18** was 300 nm or more, no cracks occurred in the covering layer **38** in all four samples, and the crack occurrence rate was 0%.

[0043] In this experiment, cracks were observed in a case of a thickness of less than 200 nm (0.2

μm), where an increase in compressive stress becomes significant in the above simulation, and no cracks were observed in a case of a thickness of 200 nm or more, where the increase in compressive stress is relatively gradual in the above simulation. Thus, the results of the present experiment correspond to the results of the above simulation.

Spacer

[0044] The inventor of the present application finds from the results of the above-mentioned simulation and experiment that the occurrence of damage to the covering layer **38** can be suppressed depending on the thickness of the adhesive **18** between the covering layer **38** on the wiring **36** and the first substrate **12**. Thus, in the present embodiment, the spacer **40** is provided between the first substrate **12** and the second substrate, and in a case of bonding, the spacer **40** causes the adhesive **18** between the covering layer **38** on the wiring **36** and the first substrate **12** to have a thickness large enough to suppress the occurrence of damage to the covering layer **38**. In the present embodiment, the thickness large enough to suppress the occurrence of damage to the covering layer **38** on the wiring **36** is based on a load in a case of bonding and stress due to warpage of the substrate during the manufacturing step and expansion and contraction of the sealing agent.

Overview of Spacer

[0045] A plurality of the spacers **40** are formed on at least one of the one surface **12a** of the first substrate **12** and the one surface **14a** of the second substrate **14**. That is, all of the spacers **40** may be provided on the one surface **12a** side, the one surface **14a** side, or both of the one surface **12a** side and the one surface **14a** side. It should be noted that a plurality of the spacers **40** do not necessarily have to be provided, and only one spacer **40** may be provided depending on the size of the substrate, the pattern of the wiring **36**, and the like.

[0046] The spacer **40** is provided in a position where the spacer **40** does not overlap the wiring **36** or the covering portion of the covering layer **38** that covers the wiring **36** in a plane orthogonal to the laminated direction (i.e., a plane parallel to the first substrate **12** and the second substrate **14**) in a case of bonding the first substrate **12** and the second substrate **14** together. In other words, the spacer **40** is provided in a position where the spacer **40** does not overlap the wiring **36** or the covering portion in a case where the bonded substrate **10** is viewed in the laminated direction (i.e., the upper or lower side in FIGS. 1A and 1B). In a case where the spacer **40** is provided in a position where the spacer **40** overlaps the wiring **36** and the covering portion, there is a possibility that a load in a case of bonding the first substrate **12** and the second substrate **14** together concentrates on the wiring **36** and the covering portion through the spacer **40** and that the covering portion of the covering layer **38** is damaged. Further, the length (hereinafter also referred to as “height”) of the spacer **40** in the laminated direction is greater than the sum of the thicknesses of the wiring **36** and the covering portion.

[0047] Appropriately adjusting the arranged position and the length in the laminated direction of the spacer **40** makes it possible to control the thickness of the adhesive **18** between the covering layer **38** on the wiring **36** and the bonded surface of the substrate facing the covering layer **38**. As a result, for example, even in a case where the covering layer **38** formed of a hard film such as a Si compound film is formed on the wiring **36** made of a relatively soft metal including aluminum, damage to the covering layer **38** can be suppressed. Further, even in a case where the relationship between the thickness T1 of the wiring **36** and the thickness T2 of the covering layer **38** on the wiring **36** is $T1 > T2$, damage to the covering layer **38** can be suppressed.

[0048] There is known a technique in which an adhesive containing a filler of a predetermined particle size is used as a spacer to control the thickness of an adhesive layer bonding the first substrate **12** and the second substrate **14** together without providing the first substrate **12** or the second substrate **14** with the spacer **40**. However, such a technique has a possibility that in a case of bonding the first substrate **12** and the second substrate **14** together, the filler overlaps the covering portion of the covering layer **38** that covers the wiring **36** and damages the covering layer **38**.

(covering portion).

[0049] In a case where the present invention is applied to a multilayer wiring substrate, spacers are arranged only in positions corresponding to wiring projected from a provided substrate surface and a covering layer on the wiring. Incidentally, in a case where non-convex multilayer wiring has sufficient strength, spacers may be arranged on the multilayer wiring.

Arrangement of Spacers

[0050] Next, arrangements of the spacers **40** in the first substrate **12** and the second substrate **14** will be described with reference to FIGS. 5A to 5C. FIGS. 5A to 5C are diagrams for explaining arrangements of the spacers **40** in the substrates, FIG. 5A is a diagram showing the spacers **40** formed in the second substrate **14**, FIG. 5B is a diagram showing the spacers **40** formed in the first substrate **12**, and FIG. 5C is a diagram showing the spacers **40** formed in the first substrate **12** and the second substrate **14**.

[0051] As described above, the spacer **40** may be provided in the first substrate **12** or the second substrate **14** as long as the spacer **40** does not contact the wiring **36** or the covering portion of the covering layer **38** that covers the wiring **36** (see FIGS. 5A and 5B). Alternatively, the spacer **40** may be provided on both the first substrate **12** and the second substrate **14** (see FIG. 5C).

[0052] The spacer **40** can be formed at the same time as at least one functional film provided in a substrate. In other words, the spacer **40** can be formed of the same material as the material for at least one functional film provided in the substrate and in the same process. Accordingly, in a case where a functional film is formed on the one surface **14a** side of the second substrate **14** (see FIG. 5A), the spacer **40** can be formed of the same material as that for the functional film during the step of forming the functional film. Further, in a case where a functional film is formed on the one surface **12a** side of the first substrate **12**, the spacer **40** can be formed of the same material as that for the functional film during the step of forming the functional film. The material for the spacer **40** may be different from that for the functional film. Even in a case where the spacer **40** is formed of the same material as that for the functional film, the spacer **40** may be formed in a step different from the step of forming the functional film made of the same material.

[0053] Further, in a case where the adhesion of the spacer **40** is low on the one surface **14a** side of the second substrate **14**, there is a possibility that the spacer **40** drops in the step of forming the spacer **40**. In such a case, in a case where the adhesion of the spacer **40** on the one surface **12a** side of the first substrate **12** is relatively high, the spacer **40** may be formed on the one surface **12a** side of the first substrate **12** (see FIG. 5B). Accordingly, in this case, the spacer **40** is formed directly on the one surface **14a** on which no functional film is formed. That is, the spacer **40** may be formed directly on the bonded surface of a substrate.

Height of Spacer

[0054] Next, the height of the spacer **40** will be described. FIG. 6 is a diagram showing the thickness of the adhesive **18** near the spacer **40** during bonding. It should be noted that this description uses as an example a case where the spacer **40** is formed on the one surface **14a** side of the second substrate **14**.

[0055] In a case of bonding the first substrate **12** and the second substrate **14** together, the spacer **40** contacts the one surface **12a**, so that the thickness H3 (see FIG. 6) of the adhesive **18** between the first substrate **12** and the second substrate **14** can be adjusted to the height of the spacer **40**. More specifically, the thickness H3 is the thickness of the adhesive **18** between the one surface **12a** of the first substrate **12** and a functional film (the covering layer **38** not located on the wiring **36** in FIG. 6) provided on the one surface **14a** of the second substrate **14**. Further, adjusting the height of the spacer **40** makes it possible to adjust the thickness H2 (see FIG. 6) of the adhesive **18** between the covering layer **38** on the wiring **36** and the one surface **12a**.

[0056] In a case of bonding the first substrate **12** and the second substrate **14** together, it is not necessary for all the spacers **40** to contact the one surface **12a**. That is, there may be spacers **40** with the adhesive **18** remaining between the spacers **40** and the one surface **12a** according to

variations in the heights of the spacers **40**, variations in the thicknesses of the first substrate **12** and the second substrate **14**, and the like (see FIG. **6**). In order to suppress variations in the thickness **H3**, it is preferable that the thickness **H1** (see FIG. **6**) of the adhesive **18** between the spacer **40** and the one surface **12a** be 0 μm or more and 0.3 μm or less. In the present embodiment, since the thickness **H2** (see FIG. **6**) is adjusted by the spacer **40**, at least one of the plurality of spacers **40** contacts the one surface **12a**, that is, the thickness **H1** is 0 μm .

[0057] Further, based on the findings made by the inventor of the present application, in order to suppress damage to the covering layer **38** on the wiring **36**, it is necessary to set the thickness **H2** of the adhesive **18** between the covering layer **38** on the wiring **36** and an opposing substrate to be greater than the thickness **H1** of the adhesive **18** between the spacer **40** and the opposing substrate. The thickness **H2** corresponds to the thickness of the adhesive layer (Adhesive thickness) in the table in FIG. **4**.

[0058] Thus, in view of the results of the above simulation and experiment, in order to suppress damage to the covering layer **38** on the wiring **36**, it is necessary that $H2 > H1$ and that the thickness **H2** be increased to relieve stress generated during the manufacturing step. Based on the results of the above simulation and experiment, the thickness **H2** is preferably set to $H2 \geq 0.6 \mu\text{m}$ at which the stress is reduced by 25% based on the stress in a case where the thickness **H2** at which damage occurs is set to $H2 \approx 0.2 \mu\text{m}$ (see FIG. **3B**). Although not shown in FIG. **3B**, it is more preferable that $H2 \geq 1.0 \mu\text{m}$ at which the stress is reduced by approximately 50% in the above simulation conducted by the inventor of the present application.

[0059] In a case where the thickness **H3** is increased, the flow path **30** and the piezoelectric element accommodation portion **28** cannot be cohesively broken in a case of the adhesive **18** being transferred, and the adhesive **18** remains in the openings of the flow path **30** and the piezoelectric element accommodation portion **28** or overflows. This may affect the functions of the flow path **30** and the piezoelectric element accommodation portion **28**. Thus, the thickness **H3** adjustable by the spacer **40** is preferably 3.5 μm or less. Accordingly, an upper limit value for the height of the spacer **40** is, for example, 3.5 μm or a predetermined value smaller than 3.5 μm . Further, a lower limit value for the height of the spacer **40** is a value that satisfies the above conditions for the **H2** (i.e., it is preferable that $H2 \geq 0.6 \mu\text{m}$, and it is more preferable that $H2 \geq 1.0$) in accordance with the sum **H4** of the thickness of the wiring **36** and the thickness of the covering layer **38** on the wiring **36**. Thus, in the present embodiment, the spacer **40** is configured so that the thickness of the adhesive **18** between the covering layer **38** on the wiring and the opposing substrate is a thickness that makes it difficult to cause damage to the covering layer **38** based on a load in a case of bonding and stress due to warpage of the substrate caused by temperature changes during a process or due to expansion and contraction of the sealing agent.

Shape of Substrate and Spacer

[0060] In a case where substrates are bonded together using an adhesive, a recess, such as an adhesive escape groove, may be formed on the bonded surface of a substrate to accommodate an excess adhesive. However, in a case where such a recess is formed on the bonded surface of a substrate facing the spacer **40** (the one surface **12a** in FIG. **6**), a crack may occur in the substrate with the recess formed.

[0061] Here, a description will be given of the occurrence of cracks due to the spacer **40** for a substrate in which a recess is formed with reference to FIGS. **7A** and **7B**. FIGS. **7A** and **7B** are diagrams showing the positional relationship between a recess **700** for accommodating the excess adhesive **18** and the spacer **40**. FIG. **7A** is a cross-sectional view of the vicinity of the spacer **40** in the first substrate **12** in which the recess **700** is formed and the second substrate **14** on which the spacer **40** is provided. FIG. **7B** is a diagram showing the positions of a region **S1** that can abut an upper surface **40a** of the spacer **40** on the one surface **12a** and the recess **700** near the region **S1**.

[0062] In a case where the first substrate **12** in which the recess **700** is formed on the one surface **12a** is bonded to the second substrate **14** in which the spacer **40** is provided on the one surface **14a**

side, a load generated between the spacer **40** and the one surface **12a** during bonding is applied to portions other than the recess **700** in the region **S1**. That is, in a case where the recess **700** is not formed, a load is applied to the entire surface of the region **S1**. On the other hand, in a case where the recess **700** is formed, a load is applied to a region **Sf** (see a hatched portion in FIG. 7B) in the region **S1** where the recess **700** is not formed, that is, an area smaller than the region **S1**. Thus, cracks may occur in the first substrate **12** in which the recess **700** is formed.

[0063] That is, in a case where the ratio of the total area of the recesses **700** in the region **S1** to the area of the region **S1** increases, the area of the region **Sf** in the region **S1** is reduced, which increases a load on the region **Sf** and causes a crack in the first substrate **12**. For example, in a case where the total area of the recesses **700** in the region **S1** is 50% of the area of the region **S1**, a load which is twice as much as the load in a case where no recess **700** is formed in the region **S1** is applied to the region **Sf** where no recess **700** is formed in the region **S1**.

[0064] Thus, in the present embodiment, the recesses **700** are formed so that the total area of the recesses **700** in the region **S1** is 0% or more and less than 20% of the area of the region **S1** in consideration of variations in the area of the region **S1** on the one surface **12a** that may contact the spacer **40**. It should be noted that whether a crack occurs in the first substrate **12** depends on a load generated during bonding, the material for the substrate, variations in the height of the spacer **40**, the material for the spacer **40**, and the like. Thus, the value “20%” mentioned above is determined, for example, experimentally depending on the material for substrates to be bonded together, the shape of and material for the spacer **40**, and the like.

[0065] Hereinafter, a description will be given of a technique for suppressing the occurrence of cracks due to the spacer **40** on the substrate in which the recess **700** is formed with reference to FIGS. 8A to 9C. FIGS. 8A to 8D are diagrams showing an example of the formation of the recess **700** for suppressing the occurrence of cracks. FIG. 8A shows an example of the formation of the recess **700** near the spacer **40**, and FIG. 8B is a diagram showing the positions of the region **S1** and the recess **700** near the region **S1** in FIG. 8A. FIG. 8C shows another example of the formation of the recess **700** near the spacer **40**, and FIG. 8D is a diagram showing the positions of the region **S1** and the recess **700** near the region **S1** in FIG. 8C. FIG. 9A is a diagram showing an example of the formation of the recess **700** in a case where the recess **700** is provided in the second substrate **14** on which the spacer **40** is provided. FIG. 9B is an enlarged view of the vicinity of the spacer on the second substrate **14** in FIG. 9A. FIG. 9C is a diagram showing the positions of a region **S2** overlapping the spacer **40** and the recess **700** near the region **S2** on the other surface **14b** in which the recess **700** is formed.

[0066] In order to suppress the occurrence of cracks in the first substrate **12** in which the recess **700** is formed, for example, the recess **700** is not formed in the region **S1** (that is, the area of the recess **700** in the region **S1** is 0% of the area of the region **S1**) (see FIG. 8A). In a case where there is a concern about the influence of misalignment between the first substrate **12** and the second substrate **14** during bonding, the spacer **40** may be formed in the region **S1** (see FIG. 8C). This allows the spacer **40** to be located in the region **S1** with high accuracy, so that the influence of the misalignment can be suppressed. Incidentally, in the description with reference to FIGS. 8A to 8D, a case where the area of the recess **700** in the region **S1** is 0% of the area of the region **S1** has been described, but it is only required that the area of the recess **700** in the region **S1** be 0% or more and less than 20% of the area of the region **S1**.

[0067] Further, in the step of manufacturing the bonded substrate **10** or a liquid ejection head provided with the bonded substrate **10**, stress is generated around the spacer **40** due to the difference in thermal expansion and elastic modulus between the spacer **40** and the adhesive **18**. Accordingly, in a case where the recess **700** for accommodating the excess adhesive **18** for bonding to the third substrate **16** is formed on the other surface **14b** of the second substrate **14** provided with the spacer **40** on the one surface **14a** side, the second substrate **14** may be damaged by the above-mentioned stress. Thus, in a case where the recess **700** is provided on the other surface **14b** of the

second substrate **14** provided with the spacer **40**, it is preferable that the recess **700** be arranged in a position where the recess **700** does not overlap the spacer **40** in a plane orthogonal to the laminated direction (see FIG. **9A**).

[0068] That is, the recess **700** provided on the other surface **14b** is formed in a position where the recess **700** does not overlap the spacer **40** provided on the one surface **14a** side in a plane parallel to the second substrate **14**. Alternatively, the recess **700** is not formed selectively so that the spacer **40** does not overlap the recess **700** in a plane parallel to the second substrate **14** (see FIGS. **9B** and **9C**). Incidentally, in the description with reference to FIGS. **9A** to **9C**, a case where the area of the recess **700** in the region **S1** is 0% of the area of the region **S2** has been described, but it is only required that the area of the recess **700** in the region **S2** be 0% or more and less than 20% of the area of the region **S2**.

Material for Spacer

[0069] The material for the spacer **40** is not specifically limited, but it is preferable to use a material that is not easily deformed by a load during bonding, and an inorganic material is suitable. The spacer **40** may be made of a single material or may be made of a plurality of materials. Further, it is preferable that all or a portion of the spacers **40** be formed on the second substrate **14**. In a case of forming the spacer **40** on the second substrate **14**, it is preferable that the spacer **40** be formed in a case of formation of at least one type of functional film formed on the one surface **14a** of the second substrate **14** such as the upper electrode, the piezoelectric element **26**, the lower electrode, the wiring **36**, and the pad **44**.

[0070] For example, in a case where the spacer **40** is formed at the same time as the pad **44**, since the pad **44** is often formed by Au plating, the spacer **40** is formed of relatively soft Au. In this case, in a case where a load is applied to the spacer **40** during bonding, an influence on the substrate is reduced. Incidentally, in a case where the spacer **40** is formed by Au plating, it is preferable to form a seed layer on an insulation film on the one surface **14a** of the second substrate **14**. In this case, an insulation layer on the one surface **14a** may be the same film as the covering layer **38**. Although details will be described later, for example, in a case where the covering layer **38** is SiN, TiW and Au are laminated as a seed layer on SiN by the sputtering method to form a film. Positions where the pad **44** and the spacer **40** are to be formed are then formed by resist patterning, and the Au plating is selectively grown in the positions. After that, the pad **44** and the spacer **40** are formed simultaneously by removing the excess TiW and Au.

[0071] In a case where the steps of growing the Au plating on the spacer **40** and the pad **44** are performed simultaneously as described above, the spacer **40** and the pad **44** having the same thickness are formed. However, the spacer **40** and the pad **44** having different thicknesses may be formed. For example, in a case of electrical connection, in a case where the pad **44** is thick, it is possible to easily perform the step of electrical connection and to prevent a short circuit with the substrate. In a case where the pad **44** and the spacer **40** having different thicknesses are formed, after TiW and Au are formed into a film as a seed layer, resist patterning and the step of growing the Au plating are performed on the pad **44** and the spacer **40**. In this way, the pad **44** and the spacer **40** each having a desired thickness can be formed.

Method for Manufacturing Bonded Substrate **10**

[0072] Next, a description will be given of an example of a method for manufacturing the bonded substrate **10** according to the present embodiment with reference to FIGS. **10A** to **13D**. FIGS. **10A** to **13D** are diagrams showing the step of manufacturing the bonded substrate **10**. FIGS. **10A** and **10B** are diagrams showing the step of working the first substrate **12**, and FIGS. **10C** and **10D** are diagrams showing the step of forming the functional film of the second substrate **14**. FIGS. **11A** to **11E** are diagrams showing the step of forming the functional film of the second substrate **14**. FIG. **12A** is a diagram showing the step of bonding the second substrate **14**, and FIGS. **12B** and **12C** are diagrams showing the step of working the second substrate **14**. FIGS. **13A** and **13B** are diagrams showing the step of bonding the third substrate **16**, FIG. **13C** is a diagram showing the step of

working the third substrate **16**, and FIG. **13D** is a diagram showing the resulting bonded substrate **10**. In the following description, a manufacturing method in which the spacer **40** is formed at the same time as the pad **44** will be described. Further, the method for working a substrate and the method for forming a functional film described below are not limited to the following methods, and various known techniques can be used.

Step of Working First Substrate **12**

[0073] First, the first substrate **12** is prepared, and the flow path **30**, the piezoelectric element accommodation portion **28**, a pad storage portion **1002** for storing the pad **44**, and the recess **700** are formed in the first substrate **12**. In a case where the first substrate **12** is, for example, a silicon substrate, a semiconductor manufacturing process can be used to form each of these components. Specifically, after forming a desired etching mask on the surface of the first substrate **12**, the first substrate **12** can be worked by performing Si dry etching. The etching mask can be formed by, for example, exposing, developing, and patterning a novolac-based photoresist. For Si dry etching, it is possible to use an etching technique called the so-called Bosch process using, for example, SF₆ gas in an etching step and C₄F₈ gas in a coating step. Further, it is also possible to thin the substrate as necessary.

[0074] Specifically, the above working method is used to form the flow path **30** penetrating from the one surface **12a** to the other surface **12b** of the first substrate **12**. Further, the piezoelectric element accommodation portion **28**, the pad storage portion **1002**, and the recess **700** are formed on the one surface **12a** (see FIG. **10A**). These components may be formed simultaneously in the same step, or may be formed individually in different steps. It should be noted that the recess **700** for storing the excess adhesive **18** is formed continuously around the flow path **30** and the piezoelectric element accommodation portion **28** on the one surface **12a**, but is not formed in a location corresponding to the spacer **40** provided on the second substrate **14** during bonding with the second substrate **14**.

[0075] Thereafter, the adhesive **18** is applied by transfer onto the one surface **12a** of the first substrate **12** (see FIG. **10B**). In this case, the adhesive **18** is not transferred onto the flow path **30**, the piezoelectric element accommodation portion **28**, the pad storage portion **1002**, and the recess **700**. In the present embodiment, the adhesive **18** is made of a benzocyclobutene resin, which is a thermosetting resin whose viscosity varies according to the temperature, so that the height of the spacer **40** is easily controlled during bonding. Further, the adhesive **18** is preferably formed thick to eliminate voids during bonding, and the thickness of the adhesive **18** is equal to or greater than the height of the spacer **40**. Incidentally, in a case where the thickness of the adhesive **18** is too large, the adhesive **18** is likely to overflow onto the flow path **30** and the piezoelectric element accommodation portion **28** on the one surface **12a**. Thus, in the present embodiment, the height of the spacer **40** is 2.0 μm, while the thickness of the adhesive **18** is 2.3 μm.

[0076] For the adhesive **18**, a material having properties such as high adhesion to substrates, high applicability with few air bubbles, and low viscosity for easy thinning is used. The adhesive **18** does not necessarily have to have all of these properties, but may have one or two of these properties. The properties of the adhesive **18** are not limited to the above-mentioned properties, and the adhesive **18** may have various known properties according to properties required for the bonded substrate **10**. Thus, the adhesive **18** is not limited to a benzocyclobutene resin, but may be, for example, an epoxy resin, acrylic resin, silicone resin, polyamide resin, polyimide resin, urethane resin, or the like.

[0077] In the present embodiment, the adhesive **18** is applied to the one surface **12a** by spin-coating the adhesive **18** on a dry film and transferring the adhesive **18** to the one surface **12a**. Incidentally, the method for applying the adhesive **18** to the bonded surface of the substrate is not limited to this. The application method may be, for example, screen printing, or in a case where the adhesive **18** is a photosensitive adhesive, a photolithography patterning technique may be used.

Step of Forming Functional Film in Second Substrate **14**

[0078] Next, the second substrate **14** is prepared, and the vibration film **24**, lower electrode, piezoelectric element **26**, upper electrode, wiring **36**, covering layer **38**, and the like are formed on the one surface **14a** of the second substrate **14** (see FIG. **10C**). The vibration film **24** and the insulation film **34** are formed on the one surface **14a** to form the lower electrode, piezoelectric element **26**, upper electrode, and wiring **36**. The method for forming these components is not limited to the above-mentioned method, and various known techniques can be used. For example, the wiring **36** is formed of AuCu by the sputtering method to have a thickness of $0.7\ \mu\text{m}$. The coating layer **38** is then formed of SiN to have a thickness of $0.2\ \mu\text{m}$ so as to cover the wiring **36**. After that, a portion of the covering layer **38** is removed as necessary. In the present embodiment, the covering layer **38** is patterned and removed to form an opening **1004** in a location to communicate with the pressure chamber **20** later and an opening **1006** in a location where the pad **44** and the wiring **36** are connected to each other.

[0079] A seed layer **1008** is then formed on the one surface **14a** side of the second substrate **14** on which the covering layer **38** is formed (see FIG. **10D**). Specifically, on the one surface **14a** side on which various functional films including the covering layer **38** are formed, a TiW film having a thickness of $0.2\ \mu\text{m}$ is formed by the sputtering method and a Au film having a thickness of $0.05\ \mu\text{m}$ is formed by the sputtering method to form the seed layer **1008**.

[0080] Next, the spacer **40** and the pad **44** are formed. Specifically, first, a resist **1102** is applied to the one surface **14a** side of the second substrate **14** on which the seed layer **1008** is formed, and patterning is performed so that the resist **1102** remains in areas other than the positions where the pad **44** and the spacer **40** are to be formed through exposure and development (see FIG. **11A**). The position where the pad **44** is formed includes the opening **1006**. A plating layer **1104** is then formed on the one surface **14a** side of the second substrate **14** (see FIG. **11B**). In the present embodiment, Au plating is used to form the plating layer **1104**, and a known plating solution is used to form the Au plating. The plating layer **1104** may be formed by either electrolytic plating or electroless plating, but in a case where the plating layer **1104** is thick, it is preferable to use electrolytic plating. In electrolytic plating, the second substrate **14** is immersed in a Au plating solution and a current is passed through the surface, thereby growing a plating layer in the portion where the resist **1102** has been removed by patterning. The resist **1102** is then peeled off (see FIG. **11C**).

[0081] Incidentally, in a case where the spacer **40** and the pad **44** have different thicknesses, the step of patterning with the resist **1102** and growing the Au plating, and the subsequent step of peeling off the resist are performed for each of the spacer **40** and the pad **44**. The order of formation is not particularly limited, but in consideration of the applicability and coating performance of the resist **1102**, it is preferable to form one with the thinner Au plating first.

[0082] After the resist **1102** is peeled off, the seed layer **1008** is then removed (see FIG. **11D**). Specifically, after the resist **1102** is peeled off, the seed layer **1008** exposed to the outside is etched. A solution represented by aqua regia capable of selectively etching Au is used to etch Au that forms the seed layer **1008**. Further, a hydrogen peroxide solution diluted to a concentration of 30% is used to etch TiW that forms the seed layer **1008**.

[0083] By removing the seed layer **1008** exposed to the outside by etching, the plating layer **1104** remains in the positions where the spacer **40** and the pad **44** are to be formed. The plating layer **1104** remaining in this manner forms the spacer **40** and the pad **44**. Thus, in the present embodiment, the spacer **40** and the pad **44** are formed of the seed layer **1008** and the plating layer **1104**. In the present embodiment, the height of the spacer **40** thus formed (the thickness of a combination of the seed layer **1008** and the plating layer **1104**) is $2.0\ \mu\text{m}$.

[0084] After the spacer **40** and the pad **44** are formed, the insulation film **34** exposed in the opening **1004** is then removed (see FIG. **11E**). Specifically, the insulation film **34** located in the opening **1004** provided in the location where the flow path **30** and the pressure chamber **20** communicate with each other is etched and removed. As a result, the one surface **14a** of the second substrate **14** is exposed in the opening **1004**. Step of Bonding First Substrate **12** and Second Substrate **14**

Together

[0085] Next, the first substrate **12** and the second substrate **14** are bonded together (see FIG. **12A**). Specifically, the first substrate **12** (see FIG. **10B**) worked in the above-mentioned step of forming the first substrate **12** is bonded to the second substrate **14** (see FIG. **11E**) worked in the above-mentioned step of forming the second substrate **14**. More specifically, the first substrate **12** and the second substrate **14** are aligned (registered) with the one surface **14a** side of the second substrate **14** facing the one surface **12a** side of the first substrate **12**, and the first substrate **12** and the second substrate **14** are adhered to each other under a load. In this case, the first substrate **12** and the second substrate **14** are heated to soften the adhesive **18**. In the registered state, the spacer **40** formed on the one surface **14a** of the second substrate **14** and the recess **700** formed on the one surface **12a** of the first substrate **12** facing the spacer **40** do not overlap in a plane orthogonal to the laminated direction. The adhesive **18** is then cured to bond the first substrate **12** and the second substrate **14** together. The adhesive **18** can be cured by a thermosetting method, an ultraviolet delay-curing method, or the like. In a case where either of the substrates to be bonded together has ultraviolet transparency, an ultraviolet curing method can be used.

[0086] During this bonding, the one surface **12a** of the first substrate **12** does not contact the covering layer **38** on the wiring **36** due to the spacer **40**. Further, the thickness (see H2 in FIG. **6**) of the adhesive **18** between the one surface **12a** and the covering layer **38** on the wiring **36** is also ensured to be constant by the spacer **40**. This makes it possible to suppress damage to the covering layer **38** on the wiring **36**. Specifically, since the thickness of the wiring **36** is 0.7 μm , the thickness of the covering layer **38** is 0.2 μm , and the thickness of the spacer **40** is 2.0 μm , the thickness (H2 in FIG. **6**) of the adhesive **18** between the one surface **12a** and the covering layer **38** on the wiring **36** can be ensured to be at least 1.1 μm . Further, since the thickness of the adhesive is 2.3 μm and the thickness of the spacer **40** is 2.0 μm , the thickness (H1 in FIG. **6**) of the adhesive between the one surface **12a** and the spacer **40** is 0 μm or more and less than 0.3 μm . That is, the conditions, $H2 > H1$ and $H2 \geq 1.0 \mu\text{m}$, which are obtained from the results of the above simulation and experiment conducted by the inventor of the present application, for suppressing damage to the covering layer **38** on the wiring **36**, are satisfied. Thus, in the present embodiment, the spacer **40** makes the thickness (H2 in FIG. **6**) of the adhesive **18** between the one surface **12a** and the covering layer **38** on the wiring **36** large enough to suppress damage to the covering layer **38**.

Step of Working Second Substrate **14**

[0087] After that, the opening **22** and the recess **700** for forming the pressure chamber **20** are formed in the second substrate **14**. Specifically, first, the surface of the second substrate **14** that is not bonded to the first substrate **12** is worked, and the second substrate **14** is thinned to form the other surface **14b** facing the one surface **14a** of the second substrate **14** (see FIG. **12B**). It should be noted that since various known techniques can be used to thin the second substrate **14**, the description thereof is omitted. Next, the opening **22** is formed in the thinned second substrate **14** (see FIG. **12C**). Incidentally, as described above, the spacer **40** is arranged so as not to overlap the opening **22** in which the pressure chamber **20** is formed in a plane orthogonal to the laminated direction. Further, the recess **700** for accommodating the excess adhesive **18** is formed in the thinned second substrate **14** (see FIG. **12C**). The recess **700** is also formed so as not to overlap the spacer **40** in a plane orthogonal to the laminated direction. The opening **22** is formed by performing resist patterning on the other surface **14b** and then performing silicon etching.

[0088] If necessary, a film serving as an etching stop layer may be formed on the lower surface (pressure chamber **20** side) of the vibration film **24**. In the present embodiment, the opening **22** and the recess **700** are formed after thinning the second substrate **14** bonded to the first substrate **12**, but the present invention is not limited to this. For example, the second substrate **14** in which the opening **22** and the recess **700** are formed in advance may be prepared and bonded to the first substrate **12**. Alternatively, the second substrate **14** that has been thinned in advance may be used to form the opening **22** and the recess **700** after being bonded to the first substrate **12**.

Step of Bonding Third Substrate

[0089] Next, the third substrate **16** is bonded to the other surface **14b** of the second substrate **14**. Specifically, first, the adhesive **18** is transferred and applied to the other surface **14b** of the second substrate **14** (see FIG. **13A**). In this case, the adhesive **18** is not transferred to the opening **22** and the recess **700**. To the method for applying the adhesive **18** used in this case, various adhesives and application methods as described above for transferring the adhesive **18** to the one surface **12a** of the first substrate **12** can be applied.

[0090] Thereafter, the third substrate **16** is prepared, and with the one surface **16a** side of the third substrate **16** facing the other surface **14b** of the second substrate **14**, the second substrate **14** and the third substrate **16** are registered and adhered to each other under a load. In this case, the second substrate **14** and the third substrate **16** are heated to soften the adhesive **18**. The adhesive **18** is then cured to bond the third substrate to the second substrate **14**. The method for curing the adhesive **18** is appropriately changed depending on the type of adhesive used.

Step of Working Third Substrate

[0091] Next, the ejection port **32** is formed in the third substrate **16**. Specifically, first, the surface of the third substrate **16** that is not bonded to the second substrate **14** is worked, and the third substrate **16** is thinned to form the other surface **16b** facing the one surface **16a** of the third substrate **16**. It should be noted that since various known techniques can be used to thin the third substrate **16**, the description thereof is omitted. Next, the ejection port **32** is formed in the thinned third substrate **16** (see FIG. **13C**). The ejection port **32** is formed by performing resist patterning on the other surface **16b** and performing silicon etching.

[0092] In the present embodiment, after the third substrate **16** is bonded to the second substrate **14**, the ejection port **32** is formed after the third substrate **16** is thinned, but the present invention is not limited to this. For example, the third substrate **16** in which the ejection port **32** is formed in advance may be prepared to bond the third substrate **16** to the second substrate **14**. Alternatively, the third substrate **16** that has been thinned in advance may be used to form the ejection port **32** after being bonded to the second substrate **14**. After that, in the step of division into chips that can be mounted on a liquid ejection head, a removal portion **1302** (see FIG. **13C**) located on the pad storage portion **1002** in the first substrate **12** is removed, so that the bonded substrate **10** can be obtained (see FIG. **13D**).

Example of Applying Bonded Substrate **10**

[0093] Next, a description will be given of a liquid ejection head including the bonded substrate **10** according to the present embodiment and a liquid ejection apparatus including the liquid ejection head. As an example of the liquid ejection apparatus, a printing apparatus that ejects ink as liquid to perform printing on a conveyed print medium will be described. FIG. **14A** is a schematic configuration diagram of the printing apparatus. FIG. **14B** is a perspective configuration diagram of a print head as the liquid ejection head including the bonded substrate **10** in the printing apparatus in FIG. **14A**.

[0094] A printing apparatus **1400** in FIGS. **14A** and **14B** is a so-called full-line type printing apparatus and includes a long print head **1402** extending across the entire width of a print medium **M** (see FIG. **14A**). The printing apparatus **1400** also includes a conveying portion **1404** that conveys the print medium **M** in a direction intersecting (in the present embodiment, orthogonal to) the width direction. In the printing apparatus **1400**, an image is printed on the print medium **M** by ejecting ink from the print head **1402** while conveying the print medium **M** by the conveying portion **1404**. In the present embodiment, the print head **1402** is configured to be able to eject four inks, cyan ink, magenta ink, yellow ink, and black ink. As a result, the printing apparatus **1400** is configured to be able to print a color image.

[0095] On a surface, facing the conveyed print medium **M**, of the print head **1402**, a plurality of the bonded substrates **10** serving as print element substrates are arrayed in a direction in which the print head **1402** extends (see FIG. **14B**). The print head **1402** also includes a liquid supply unit

1406 in which a circulation flow path for supplying ink supplied from an ink tank (not shown) to each bonded substrate **10** and collecting ink that has not been ejected from the ejection port **32** during printing from the bonded substrate **10** is formed. The print head **1402** also includes a negative pressure control unit **1410** for controlling pressure (negative pressure) in the circulation flow path, and a liquid connection portion **1412** serving as a port through which ink is supplied to and discharged from the liquid supply unit **1406** (see FIG. **14A**).

[0096] Further, the print head **1402** is electrically connected to an electric control portion that supplies power and an ejection control signal to the print head **1402**. Specifically, each bonded substrate **10** is connected to the same electric wiring substrate **1416** via a flexible wiring substrate **1414**. The electric wiring substrate **1416** is provided with a power supply terminal **1418** for receiving power and a signal input terminal **1420** for receiving an ejection control signal. A print element (piezoelectric element **26**) provided in the bonded substrate **10** is driven using power supplied from the power supply terminal **1418** based on the ejection control signal input from the signal input terminal **1420**. By driving the print element in this way, ink supplied from the liquid supply unit **1406** and stored in the pressure chamber **20** is ejected from the ejection port **32** in each bonded substrate **10**.

Function and Effect

[0097] As described above, the bonded substrate **10**, which is a liquid ejection head substrate, is provided with the spacer **40** between the one surface **14a** of the second substrate **14** on which the wiring **36** is formed and the one surface **12a** of the first substrate **12** bonded to the one surface **14a** via the adhesive **18**. The spacer **40** is arranged in a position where the spacer **40** does not contact the wiring **36** or the covering layer **38** on the wiring **36**. The spacer **40** is also arranged so that in a case of bonding substrates together, the thickness $H2$ of the adhesive **18** between the covering layer **38** on the wiring **36** and the bonded surface of an opposite substrate is a value that makes it difficult for the covering layer **38** to be damaged by a load during bonding and stress due to warpage of the substrate caused by temperature changes and expansion and contraction of the sealing agent. Specifically, the height of the spacer **40** is set so that the thickness $H2$ is greater than the thickness $H1$ of the adhesive **18** between the spacer **40** and the bonded surface of the opposite substrate, and $H2 \geq 0.6 \mu\text{m}$, more preferably $H2 \geq 1.0 \mu\text{m}$.

[0098] As a result, in the bonded substrate **10**, damage to the covering layer **38** on the wiring **36** due to a load generated during bonding between the first substrate **12** and the second substrate **14** is suppressed. Further, in the step of manufacturing the bonded substrate **10** and the step of manufacturing a liquid ejection head using the bonded substrate **10**, damage to the covering layer **38** on the wiring **36** due to stress caused by warpage of the substrate due to temperature changes and expansion and contraction of the sealing agent is suppressed.

Other Embodiments

[0099] The above-described embodiment may be modified as shown in (1) to (4) below. [0100] (1) In the above embodiment, the spacer **40** is provided so as not to overlap the wiring **36** and the covering layer **38** on the wiring **36** in a plane orthogonal to the direction in which substrates are laminated, and $H2 > H1$ and $H2 \geq 0.6$ are satisfied, but the present invention is not limited to this. For example, in a case where stress due to warpage caused by changes in the temperature of a substrate used and expansion and contraction of the sealing agent is small, it is only required that the height of the spacer **40** be greater than the sum of the thicknesses of the wiring **36** and the covering layer **38** on the wiring **36**, that is, $H2 > H1$ be satisfied. [0101] (2) In the above embodiment, the piezoelectric element **26** is provided as an energy generation element, but the present invention is not limited to this. Various known energy generation elements, such as an electrothermal conversion element, may be used as the energy generation element. Further, in the above embodiment, the recess **700** is hole-shaped, but the shape of the recess **700** is not limited to this. The recess **700** may have a groove shape extending in a predetermined direction on a surface of the substrate. Furthermore, in the above embodiment, the full-line type print head is used as a liquid

ejection head, but the liquid ejection head is not limited to this, and a so-called serial scan type print head may be used. [0102] (3) Although not specifically described in the above embodiment, the spacer **40** has a substantially cubic shape as shown in FIGS. 7A and 7B, but the shape is not limited to this and may be a columnar shape such as a cylindrical shape, an elliptical cylindrical shape, and a polygonal cylindrical shape. Further, the spacer **40** does not necessarily have to have a columnar shape and may have a substantially conical shape whose width gradually increases toward an opposite substrate. [0103] (4) The above embodiment and the various modifications shown in (1) to (3) above may be combined as appropriate.

[0104] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0105] This application claims the benefit of Japanese Patent Application No. 2024-031791, filed Mar. 4, 2024, which is hereby incorporated by reference wherein in its entirety.

Claims

1. A liquid ejection head substrate comprising: an energy generation element for imparting energy to liquid, the liquid ejection head substrate comprising: a first substrate; a second substrate comprising a bonded surface bonded to a bonded surface of the first substrate via an adhesive and provided with wiring connected to the energy generation element and a covering layer covering the wiring; and a member provided on the bonded surface of at least one of the first substrate and the second substrate and in a position where the member does not overlap the wiring in a direction in which the first substrate and the second substrate are laminated, wherein the member has a height so that a second thickness of the adhesive on the covering layer provided on the wiring is greater than a first thickness of the adhesive on the member.
2. The liquid ejection head substrate according to claim 1, wherein the member has a height large enough to ensure, as the second thickness, a thickness capable of suppressing occurrence of damage to the covering layer on the wiring based on a load during the bonding and stress due to warpage of a substrate caused by a temperature change and due to expansion and contraction of a sealing agent sealing a connection portion connected with an external terminal.
3. The liquid ejection head substrate according to claim 2, wherein the second thickness is 0.6 μm or more.
4. The liquid ejection head substrate according to claim 3, wherein the second thickness is 1.0 μm or more.
5. The liquid ejection head substrate according to claim 1, wherein the first thickness is 0 μm or more and less than 0.3 μm .
6. The liquid ejection head substrate according to claim 1, wherein a third thickness of the adhesive between the first substrate and the second substrate during the bonding is 3.5 μm or less.
7. The liquid ejection head substrate according to claim 1, wherein a first recess capable of accommodating the excess adhesive is formed on the bonded surface of the first substrate.
8. The liquid ejection head substrate according to claim 7, wherein the member is provided on the bonded surface of the second substrate, and the first recess is formed in a region which the member may contact during the bonding so that an area of the first recess formed in the region is 0% or more and less than 20% of an area of the region.
9. The liquid ejection head substrate according to claim 7, wherein the member is provided on the bonded surface of the first substrate, and the first recess is formed in a region where the member is provided on the bonded surface so that an area of the first recess formed in the region is 0% or more and less than 20% of an area of the region.
10. The liquid ejection head substrate according to claim 1, wherein a liquid chamber storing liquid

to which energy is to be imparted by the energy generation element is formed in the second substrate, and the member is provided in a position where the member does not overlap the liquid chamber in a plane orthogonal to the laminated direction.

11. The liquid ejection head substrate according to claim 10, further comprising a third substrate bonded to a second bonded surface of the second substrate facing a first bonded surface bonded to the first substrate via the adhesive and comprising an ejection port communicating with the liquid chamber and being capable of ejecting the liquid stored in the liquid chamber, wherein a second recess capable of accommodating the excess adhesive is formed on the second bonded surface, and the second recess is formed in a region where the second recess overlaps the member during the bonding in a plane parallel to the second substrate so that an area of the second recess formed in the region is 0% or more and less than 20% of an area of the region.

12. The liquid ejection head substrate according to claim 1, wherein the member is formed of an inorganic material.

13. The liquid ejection head substrate according to claim 1, wherein the member is provided on the bonded surface of the second substrate and formed of a material identical to a material for at least one functional film formed in the second substrate.

14. The liquid ejection head substrate according to claim 13, wherein the member is formed of a material identical to a material for a pad for electrical connection.

15. The liquid ejection head substrate according to claim 14, wherein the member is formed to have a thickness equal to a thickness of the pad.

16. The liquid ejection head substrate according to claim 14, wherein the member is formed to have a thickness different from a thickness of the pad.

17. The liquid ejection head substrate according to claim 1, wherein the wiring is formed of metal containing at least aluminum, and the covering layer is formed of a silicon compound film.

18. The liquid ejection head substrate according to claim 1, wherein the wiring has a thickness greater than a thickness of the covering layer.

19. The liquid ejection head substrate according to claim 1, wherein the member is formed of metal made of a seed layer and metal made of a plating layer.

20. A method for manufacturing a liquid ejection head substrate comprising an energy generation element for imparting energy to liquid, the method comprising: preparing a first substrate; preparing a second substrate comprising a bonded surface bonded to a bonded surface of the first substrate via an adhesive and provided with wiring connected to the energy generation element and a covering layer covering the wiring; and bonding the first substrate and the second substrate together via the adhesive, wherein on the bonded surface of at least one of the first substrate and the second substrate, a member is provided in a position where the member does not overlap the wiring in a direction in which the first substrate and the second substrate are laminated, wherein in the bonding, the first substrate and the second substrate are bonded together in a state where each of the first substrate and the second substrate is formed on a wafer, and wherein the member has a height so that a second thickness of the adhesive on the covering layer provided on the wiring is greater than a first thickness of the adhesive on the member.
